

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Water treatment, conservation & new sources - Set 1

Mark scheme

AQA spec 3.2.2.4-3.2.2.5; The physical environment; textbook pp. 122-127

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from:</i> • Topography: a narrow exit from a large deep basin so a small dam holds a large volume • Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity • Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) • Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) • Infrastructure / proximity to demand: access for construction, cost of transporting water • Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis	4
01.2	• Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities • Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation	3
01.3	• Very energy-intensive and expensive (reverse osmosis / distillation) • Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East)	2
01.4	• Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
02.1	• $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%)	1

Question	Marking guidance	Marks
02.2	• A fixed fee gives no incentive to save water because the bill is the same however much is used • Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances)	2
02.3	<i>Any 3 from:</i> • Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers • Greywater re-use: bath / shower water used to flush toilets or water gardens • Rainwater collection / water butts for garden use • Xeriscaping: planting drought-adapted species that need no watering • Repairing leaks / dripping taps	3
02.4	• Old infrastructure / pipes in poor condition leak • Repair and good maintenance / pipe replacement / leak detection reduce wastage	2
02.5	• Water is delivered directly to the plant roots / soil around each plant • Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
03.1	• Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended • Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges • The particles clump together (flocs) and settle in the clarifier tank	3
03.2	• Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources • Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste	2
03.3	• Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) • Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution)	2
03.4	• Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) • Reverse osmosis / distillation (desalination) - expensive, energy-intensive • Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
04.1	• B - rivers are accessible over a large area with few natural contaminants	1
04.2	• Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals	2

Question	Marking guidance	Marks
04.3	• Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
05.1	• Its total annual discharge	1
05.2	• Public supplies cannot match consumer demand / the water is often cleaner than rivers • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge	2
05.3	• The reservoir is created in the estuary so no farmland / settlements need to be flooded • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	• Potable: no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints • Spray irrigation: low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food • Condenser cooling: no gross solids - pipe blockage • Boiler water: no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes	4
06.2	• Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter • Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed	2
06.3	• Clay particles carry electrostatic charges that make them repel each other so they stay suspended • Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier	2
06.4	<i>Any 2 from:</i> • Chlorine • Ozone • Ultraviolet light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
07.1	• No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m • Reduction = 2.2 m	2
07.2	• After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream • During dry periods the dam is opened to release water and maintain river levels	2
07.3	• Deforestation / urbanisation reduce infiltration and interception • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods	2

Question	Marking guidance	Marks
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	